

ANN EXPIRIMENT 9

AIM: To implement Self organising maps for Clustering problem

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(1)

X = np.vstack((
    np.random.randn(100,2)+2,
    np.random.randn(100,2)-2
))

g = 10
lr = 0.5
epochs = 100
W = np.random.rand(g,g,2)
for e in range(epochs):
    for x in X:
        d = np.linalg.norm(W - x , axis=2)
        bmu = np.unravel_index(np.argmin(d), d.shape)
        for i in range(g):
            for j in range(g):
                W[i,j] += lr * (x - W[i,j])
    lr *= 0.92
clusters = []
for x in X:
    d = np.linalg.norm(W - x , axis=2)
    bmu = np.unravel_index(np.argmin(d), d.shape)
    clusters.append(bmu[0]*g + bmu[1])
clusters = np.array(clusters)
plt.figure()
plt.scatter(X[:,0],X[:,1])
```

```

plt.title("Input Data")

plt.figure()

plt.scatter(X[:,0],X[:,1],c=clusters)

plt.title("SOM Clustering")

plt.figure()

u = np.linalg.norm(W - W.mean(axis=(0,1)), axis=2)

plt.imshow(u,cmap='gray')

plt.colorbar()

plt.title("Distance Map")

plt.show()

```

OUTPUT:

